

1. Evaluate $\int_S \hat{n} \cdot \text{curl } \underline{F} \, ds$ where $\underline{F} = (2x-y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$ and S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$.

Sol:

surface $S: x^2 + y^2 + z^2 = 1$ is sphere, center $= (0,0,0)$, radius

1. In xy -plane, the boundary C : circle, center $(0,0)$, radius

1. let $x = \cos t$, $y = \sin t$, $z = 0$ and $0 \leq t \leq 2\pi$ on C .

$$d\underline{r} = dx\hat{i} + dy\hat{j} + dz\hat{k} = -\sin t \, dt \hat{i} + \cos t \, dt \hat{j}.$$

$$\underline{F} = (2x-y)\hat{i} - yz^2\hat{j} - y^2z\hat{k} = (2\cos t - \sin t)\hat{i} + 0\hat{j} + 0\hat{k}.$$

$$\underline{F} \cdot d\underline{r} = [(2\cos t - \sin t)\hat{i}] \cdot [-\sin t \, dt \hat{i} + \cos t \, dt \hat{j}].$$

$$= (-2\sin t \cos t + \sin^2 t) \, dt.$$

$$= (-\sin 2t + \frac{1}{2}(1 - \cos 2t)) \, dt.$$

$$\int_S \hat{n} \cdot \text{curl } \underline{F} \, ds = \oint_C \underline{F} \cdot d\underline{r} = \oint_C \underline{F} \cdot d\underline{r}$$

$$= \int_0^{2\pi} \left[-\sin 2t + \frac{1}{2} - \frac{1}{2} \cos 2t \right] dt$$

$$= \left[\frac{\cos 2t}{2} + \frac{t}{2} - \frac{1}{4} \frac{\sin 2t}{2} \right]_0^{2\pi}$$

$$= \left(\frac{1}{2} \cos 4\pi + \frac{2\pi}{2} - \frac{1}{4} \sin 4\pi \right) - \left(\frac{1}{2} \cos 0 + \frac{0}{2} - \frac{1}{4} \sin 0 \right).$$

$$= \frac{1}{2} + \pi - \frac{1}{2} = \pi \quad \times$$

A particle of mass m , which at time $t = 0$ has position x_0 and velocity v_0 , is being acted upon by a sinusoidal force $F = F_0 \sin(\omega t - \phi)$ where F_0 , ω and ϕ are constants. Find the equations for position and velocity for all positive time t .

Sol:-

$$\text{Let } F = F_0 \sin(\omega t - \phi).$$

$$m \frac{d^2 x}{dt^2} = F_0 \sin(\omega t - \phi)$$

$$\frac{d^2 x}{dt^2} = \frac{F_0}{m} \sin(\omega t - \phi).$$

$$v(t) = \frac{dx}{dt} = \int \frac{F_0}{m} \sin(\omega t - \phi) dt$$

$$= -\frac{F_0}{m\omega} \cos(\omega t - \phi) + C_1$$

$$\text{Since } v(t=0) = v_0,$$

$$-\frac{F_0}{m\omega} \cos \phi + C_1 = v_0 \Rightarrow C_1 = v_0 + \frac{F_0}{m\omega} \cos \phi.$$

$$\text{The velocity eqn is } v(t) = -\frac{F_0}{m\omega} \cos(\omega t - \phi) + v_0 + \frac{F_0}{m\omega} \cos \phi.$$

$$x(t) = \int \left[-\frac{F_0}{m\omega} \cos(\omega t - \phi) + v_0 + \frac{F_0}{m\omega} \cos \phi \right] dt.$$

$$= -\frac{F_0}{m\omega^2} \sin(\omega t - \phi) + \left(v_0 + \frac{F_0}{m\omega} \cos \phi \right) t + C_2$$

$$\text{Since } x(t=0) = x_0,$$

$$-\frac{F_0}{m\omega^2} \sin(-\phi) + C_2 = x_0 \Rightarrow C_2 = x_0 - \frac{F_0}{m\omega^2} \sin \phi.$$

The position equation is.

$$x(t) = -\frac{F_0}{m\omega^2} \sin(\omega t - \phi) + \left(v_0 + \frac{F_0}{m\omega} \cos \phi \right) t + x_0 - \frac{F_0}{m\omega^2} \sin \phi$$

③ If $\underline{F} = 3xy\hat{i} - y^2\hat{j}$, evaluate

$\int_C \underline{F} \cdot d\underline{r}$ where C is the curve in the xy -plane, $y = 2x^2$, from $(0,0)$ to $(1,2)$.

Solⁿ

$$\underline{F} = 3xy\hat{i} - y^2\hat{j}, \quad \underline{r} = x\hat{i} + y\hat{j}$$

$$y = 2x^2, \text{ from } (0,0) \text{ to } (1,2)$$

$$\int_C \underline{F} \cdot d\underline{r} = \int_C (3xy\hat{i} - y^2\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$= \int_C 3xy \, dx - \int_C y^2 \, dy$$

$$= \int_C 3x(2x^2) \, dx - \int_C (2x^2)^2 \, d(2x^2)$$

$$= \int_C 6x^3 \, dx - \int_C 4x^4 \, d(2x^2) \quad \text{--- (1)}$$

$$dy = d(2x^2)$$

$$\frac{dy}{dx} = 4x$$

$$dy = 4x \, dx = d(2x^2)$$

From eqⁿ (1)

$$\int_C \underline{F} \cdot d\underline{r} = \int_0^1 6x^3 \, dx - \int_0^1 4x^4 \cdot 4x \, dx$$

$$= \left. \frac{6x^4}{4} \right|_0^1 - \left. \frac{16x^6}{6} \right|_0^1$$

$$= \frac{3}{2} (1-0) - \frac{8}{3} (1-0)$$

$$= \frac{3}{2} - \frac{8}{3} = \frac{9-16}{6} = -\frac{7}{6}$$

④ (a) Show that $\underline{F} = 3(2xy + z^3)\hat{i} + 3xz^2\hat{j} + 9xz^2\hat{k}$ is a conservative force field.

(b) Find the potential energy V .

(c) Find the work done in moving an object in the force field from $(1, -2, 1)$ to $(3, 1, 4)$.

Solⁿ

$$\underline{F} = (6xy + 3z^3)\hat{i} + 3xz^2\hat{j} + 9xz^2\hat{k}$$

$$\nabla \times \underline{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy + 3z^3 & 3xz^2 & 9xz^2 \end{vmatrix}$$

$$= \left(\frac{\partial}{\partial y} 9xz^2 - \frac{\partial}{\partial z} 3xz^2 \right) \hat{i} -$$

$$\left(\frac{\partial}{\partial x} 9xz^2 - \frac{\partial}{\partial z} (6xy + 3z^3) \right) \hat{j} +$$

$$\left(\frac{\partial}{\partial x} 3xz^2 - \frac{\partial}{\partial y} (6xy + 3z^3) \right) \hat{k}$$

$$= 0\hat{i} - (9z^2 - 9z^2)\hat{j} + (6xz - 6xz)\hat{k}$$

$$= 0$$

$\therefore \underline{F}$ is conservative force.

$$(b) \underline{F} = -\nabla V = -\left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right)$$

$$-\frac{\partial V}{\partial x} = 6xy + 3z^3, \quad V = -\int (6xy + 3z^3) \, dx$$

$$V = -3x^2y + 3xz^3 + C$$

$$-\frac{\partial V}{\partial y} = 3x^2, \quad V = -\int 3x^2 \, dy = -3x^2y$$

$$-\frac{\partial V}{\partial z} = 9xz^2, \quad V = -\int 9xz^2 \, dz = -3xz^3 + C$$

$$\therefore V(x, y, z) = -3x^2y - 3xz^3 + C$$

(c) potential function $\phi(x, y, z) = -3x^2y - 3xz^3$ from $(1, -2, 1)$ to $(3, 1, 4)$.

work done, $W = \phi(3, 1, 4) - \phi(1, -2, 1)$

$$= [-3 \times 3^2 \times 1 - 3 \times 3 \times 4^3] - [-3 \times 1^2 \times (-2) - 3 \times 1 \times 1^3]$$

$$= [-27 - 144] - [-6 - 3] = -171 + 9 = -162$$

Find the gradient of the function $\phi(x, y, z) = 2x^2 - 3xy + 9z - 2$ at the point $(1, 0, 0)$ and differentiate it in the direction specified by $[2, -2, 1]$.

Sol:

$$\text{Let } \phi(x, y, z) = 2x^2 - 3xy + 9z - 2.$$

$$\nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}.$$

$$\begin{aligned} &= \frac{\partial}{\partial x} (2x^2 - 3xy + 9z - 2) \hat{i} + \\ &\quad \frac{\partial}{\partial y} (2x^2 - 3xy + 9z - 2) \hat{j} + \\ &\quad \frac{\partial}{\partial z} (2x^2 - 3xy + 9z - 2) \hat{k} \\ &= (4x - 3y) \hat{i} + (-3x) \hat{j} + 9 \hat{k} \\ &= (4x - 3y) \hat{i} - 3x \hat{j} + 9 \hat{k} \end{aligned}$$

At $(1, 0, 0)$

$$\begin{aligned} \nabla \phi &= (4 \times 1 - 3 \times 0) \hat{i} - 3(1) \hat{j} + 9 \hat{k} \\ &= 4 \hat{i} - 3 \hat{j} + 9 \hat{k} \end{aligned}$$

The unit vector in the direction $[2, -2, 1]$ is

$$\hat{a} = \frac{\underline{a}}{|\underline{a}|} \quad (1)$$

$$\underline{a} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$\begin{aligned} |\underline{a}| &= (2^2 + (-2)^2 + 1^2)^{1/2} \\ &= (4 + 4 + 1)^{1/2} = \sqrt{9} = 3. \end{aligned}$$

The unit vector in the direction

$$\begin{aligned} [2, -2, 1] \text{ is } \hat{a} &= \frac{2\hat{i} - 2\hat{j} + \hat{k}}{3} \\ &= \frac{2}{3} \hat{i} - \frac{2}{3} \hat{j} + \frac{1}{3} \hat{k} \end{aligned}$$

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The required directional derivative at $(1, 0, 0)$ is

$$\begin{aligned} \hat{a} \cdot \nabla \phi &= \left(\frac{2}{3} \hat{i} - \frac{2}{3} \hat{j} + \frac{1}{3} \hat{k} \right) \cdot (4\hat{i} - 3\hat{j} + 9\hat{k}) \\ &= \frac{8}{3} + 2 + 3 = \frac{8+6+9}{3} = \frac{23}{3} \end{aligned}$$

② If C is a closed circuit containing an area S , prove $\oint_C \phi \nabla \phi \cdot d\underline{s} = 0$.
proof:

By Stokes' Theorem,

$$\oint_C \underline{A} \cdot d\underline{s} = \int_S (\nabla \times \underline{A}) \cdot \hat{n} dS$$

$$\text{Let } \underline{A} = \phi \nabla \phi$$

By vector identity

$$\nabla \times (f \underline{v}) = f(\nabla \times \underline{v}) + (\nabla f \times \underline{v})$$

$$\nabla \times (\phi \nabla \phi) = \phi(\nabla \times \nabla \phi) + (\nabla \phi) \times (\nabla \phi)$$

$$\nabla \times \nabla \phi = 0$$

$$(\nabla \phi) \times (\nabla \phi) = 0.$$

$$\therefore \nabla \times (\phi \nabla \phi) = \phi(0) + 0 = 0.$$

$$\therefore \oint_C \phi \nabla \phi \cdot d\underline{s} = \int_S 0 \cdot \hat{n} dS = 0.$$

$$\therefore \oint_C \phi \nabla \phi \cdot d\underline{s} = 0.$$

③ Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\hat{i} - \hat{j} - 2\hat{k}$.

Sol:-

$$\phi = x^2yz + 4xz^2 \text{ at } (1, -2, -1)$$

$$\begin{aligned} \nabla \phi &= \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \\ &= (2xyz + 4z^2) \hat{i} + x^2z \hat{j} + (x^2y + 8xz) \hat{k} \end{aligned}$$

$$\text{At } (1, -2, -1)$$

$$\begin{aligned} \nabla \phi &= (2 \times 1 \times (-2) \times (-1) + 4(-1)^2) \hat{i} + (1^2 \times (-1)) \hat{j} + (1^2(-2) + 8 \times 1 \times (-1)) \hat{k} \\ &= 8\hat{i} - \hat{j} - 10\hat{k} \end{aligned}$$

The unit vector in the direction

$$\text{of } \underline{a} = 2\hat{i} - \hat{j} - 2\hat{k}$$

$$\begin{aligned} \hat{a} &= \frac{\underline{a}}{|\underline{a}|} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{(2^2 + (-1)^2 + (-2)^2)^{1/2}} \\ &= \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{9}} = \frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k} \end{aligned}$$

$$\begin{aligned} \hat{a} \cdot \nabla \phi &= \left(\frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}\right) \cdot (8\hat{i} - \hat{j} - 10\hat{k}) \\ &= \frac{16}{3} + \frac{1}{3} + \frac{20}{3} = \frac{37}{3} \end{aligned}$$

④ If C is a closed circuit containing an area S , prove

$$\oint_C \hat{n} \cdot \text{curl } \underline{E} \, ds = \oint_C \phi \underline{E} \cdot d\underline{s} - \int_S (\nabla \phi \times \underline{E}) \cdot \hat{n} \, ds.$$

Proof:

$$\oint_C \underline{A} \cdot d\underline{s} = \int_S (\nabla \times \underline{A}) \cdot \hat{n} \, ds$$

Let $\underline{A} = \phi \underline{E}$ (ϕ is scalar field and $\underline{E}$ is a vector field)

$$\nabla \times (\phi \underline{E}) = \phi (\nabla \times \underline{E}) + (\nabla \phi) \times \underline{E}$$

$$\oint_C \phi \underline{E} \cdot d\underline{s} = \int_S (\text{curl}(\phi \underline{E})) \cdot \hat{n} \, ds$$

$$= \int_S [\phi \text{curl } \underline{E} + (\nabla \phi \times \underline{E})] \cdot \hat{n} \, ds$$

$$= \int_S \phi \text{curl } \underline{E} \cdot \hat{n} \, ds + \int_S (\nabla \phi \times \underline{E}) \cdot \hat{n} \, ds$$

$$= \int_S \phi \hat{n} \cdot \text{curl } \underline{E} \, ds + \int_S (\nabla \phi \times \underline{E}) \cdot \hat{n} \, ds$$

$$\therefore \int_S \phi \hat{n} \cdot \text{curl } \underline{E} \, ds =$$

$$\oint_C \phi \underline{E} \cdot d\underline{s} - \int_S (\nabla \phi \times \underline{E}) \cdot \hat{n} \, ds$$

Exa. 1 Sol: $\underline{a} = 2\hat{i} + 2\hat{j} - \hat{k}$
 $\underline{b} = 6\hat{i} - 3\hat{j} + 2\hat{k}$

$$\underline{a} \cdot \underline{b} = ab \cos \theta$$

where θ is the angle between $\underline{a}$ and $\underline{b}$.

$$\cos \theta = \frac{\underline{a} \cdot \underline{b}}{ab}$$

$$\underline{a} \cdot \underline{b} = (2\hat{i} + 2\hat{j} - \hat{k}) \cdot (6\hat{i} - 3\hat{j} + 2\hat{k})$$

$$= 12 - 6 - 2 = 4$$

$$|\underline{a}| = a = (2^2 + 2^2 + (-1)^2)^{1/2} = \sqrt{9} = 3$$

$$|\underline{b}| = b = (36 + (-3)^2 + 2^2)^{1/2} = \sqrt{49} = 7$$

$$\cos \theta = \frac{4}{3 \times 7} = \frac{4}{21}$$

$$\theta = \cos^{-1} \frac{4}{21} = 79^\circ$$

S.A (1.1) 2 (i) Sol: $\underline{a} = \hat{i} + 2\hat{j} - 3\hat{k}$
 $\underline{b} = 2\hat{i} - \hat{j} + \hat{k}$

$$\underline{b} \times \underline{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ -1 & 4 & -1 \end{vmatrix} = \hat{i}(-1-4) - \hat{j}(-2+1) + \hat{k}(8-1)$$

$$= -3\hat{i} + \hat{j} + 7\hat{k}$$

$$\underline{a} \cdot (\underline{b} \times \underline{c}) = (\hat{i} + 2\hat{j} - 3\hat{k}) \cdot (-3\hat{i} + \hat{j} + 7\hat{k})$$

$$= -3 + 2 - 21 = -22$$

(ii) $\underline{a} \times \underline{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & -1 & 1 \end{vmatrix} = \hat{i}(2-5) - \hat{j}(1-6) + \hat{k}(-1-4)$

$$= -3\hat{i} + 5\hat{j} - 5\hat{k}$$

$$(\underline{a} \times \underline{b}) \times \underline{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 5 & -5 \\ -1 & 4 & -1 \end{vmatrix} = \hat{i}(5-25) - \hat{j}(3-5) + \hat{k}(-12+5)$$

$$= -20\hat{i} + 2\hat{j} - 7\hat{k}$$

Ex (1.2) Sol: $\underline{a} = 3\hat{i} - \hat{j} + 2\hat{k}$
 $\underline{b} = 2\hat{i} + \hat{j} - \hat{k}$
 $\underline{c} = \hat{i} - 2\hat{j} + 2\hat{k}$

$$\underline{a} \times \underline{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix}$$

$$= (1-2)\hat{i} - (-3-4)\hat{j} + (3+2)\hat{k}$$

$$= -\hat{i} + 7\hat{j} + 5\hat{k}$$

$$(\underline{a} \times \underline{b}) \times \underline{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 7 & 5 \\ 1 & -2 & 2 \end{vmatrix}$$

$$= (14+10)\hat{i} - (-2-5)\hat{j} + (2-7)\hat{k}$$

$$= 24\hat{i} + 7\hat{j} - 5\hat{k}$$

S.A (1.1) 1. Sol:

Scalar Triple product.

$$\underline{x} \cdot (\underline{y} \times \underline{z}) = (\underline{x} \times \underline{y}) \cdot \underline{z}$$

$$\text{Let } \underline{x} = \underline{a} \times \underline{b}$$

$$\text{L.H.S} = (\underline{a} \times \underline{b}) \cdot (\underline{c} \times \underline{d})$$

$$= \underline{x} \cdot (\underline{c} \times \underline{d}) =$$

$$= (\underline{x} \times \underline{c}) \cdot \underline{d}$$

$$= ((\underline{a} \times \underline{b}) \times \underline{c}) \cdot \underline{d} \quad \text{--- (1)}$$

By vector Triple product formula

$$(\underline{a} \times \underline{b}) \times \underline{c} = (\underline{a} \cdot \underline{c})\underline{b} - (\underline{b} \cdot \underline{c})\underline{a}$$

$$\text{eq (1)} ((\underline{a} \times \underline{b}) \times \underline{c}) \cdot \underline{d}$$

$$= (\underline{a} \cdot \underline{c})(\underline{b} \cdot \underline{d}) - (\underline{b} \cdot \underline{c})(\underline{a} \cdot \underline{d})$$

$$= (\underline{a} \cdot \underline{c})(\underline{b} \cdot \underline{d}) - (\underline{a} \cdot \underline{d})(\underline{b} \cdot \underline{c})$$

$$= \text{R.H.S}$$

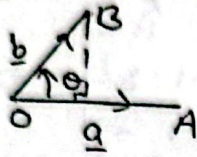
$$\therefore (\underline{a} \times \underline{b}) \cdot (\underline{c} \times \underline{d})$$

$$= (\underline{a} \cdot \underline{c})(\underline{b} \cdot \underline{d}) - (\underline{a} \cdot \underline{d})(\underline{b} \cdot \underline{c})$$

Unit. I vector Algebra. Math-2109 (UDE) Mechanics II

Dot product (or) scalar product of two vectors

Let $\underline{OA} = \underline{a}$, $\underline{OB} = \underline{b}$



$$\angle AOB = \theta$$

$$\underline{a} \cdot \underline{b} = ab \cos \theta$$

$$\begin{aligned} \underline{b} \cdot \underline{a} &= ba \cos (2\pi - \theta) \\ &= ba \cos \theta = ab \cos \theta \\ &= \underline{a} \cdot \underline{b} \end{aligned}$$

Thus scalar product is commutative.

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$$

$$\hat{i} \cdot \hat{j} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0$$

$$\underline{a} \cdot (\underline{b} + \underline{c}) = \underline{a} \cdot \underline{b} + \underline{a} \cdot \underline{c}$$

(distributive law)

eg. (1.1)

p.s

Sol:

$$\underline{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$$

$$\underline{a} \cdot \underline{a} = (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \cdot (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k})$$

$$= a_1^2 + a_2^2 + a_3^2$$

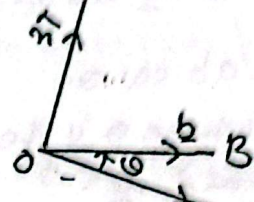
$$\underline{a} \cdot \underline{a} = aa \cos 0 = a^2 = a_1^2 + a_2^2 + a_3^2$$

$$\therefore \underline{a} \cdot \underline{a} = a^2$$

$$a = \sqrt{\underline{a} \cdot \underline{a}} = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

is the magnitude of $\underline{a}$.

cross product (or) vector product of two vectors.



$$\underline{a} \times \underline{b} = (ab \sin \theta) \hat{n}$$

$$\begin{aligned} \underline{b} \times \underline{a} &= (ba \sin \theta) (-\hat{n}) \\ &= -(\underline{a} \times \underline{b}) \end{aligned}$$

Thus vector multiplication is non-commutative.

$$\underline{a} \times \underline{a} = 0$$

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$$

$$\begin{aligned} \hat{i} \times \hat{j} &= \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j} \\ \hat{j} \times \hat{i} &= -\hat{k}, \hat{k} \times \hat{j} = -\hat{i}, \hat{i} \times \hat{k} = -\hat{j} \end{aligned}$$

$$\underline{a} \times (\underline{b} + \underline{c}) = \underline{a} \times \underline{b} + \underline{a} \times \underline{c}$$

(distributive law)

ma.2
p.10

Sol: $\underline{a} = 5\hat{i} + 3\hat{j} + 2\hat{k}$

$$\underline{b} = 2\hat{i} - \hat{j} + 3\hat{k}$$

$$\underline{c} = 7\hat{i} - 3\hat{j} + 10\hat{k}$$

$$\begin{aligned} \underline{BA} &= \underline{a} - \underline{b} = (5\hat{i} + 3\hat{j} + 2\hat{k}) - (2\hat{i} - \hat{j} + 3\hat{k}) \\ &= 3\hat{i} + 4\hat{j} - \hat{k} \end{aligned}$$

$$\begin{aligned} \underline{BC} &= \underline{c} - \underline{b} = (7\hat{i} - 3\hat{j} + 10\hat{k}) - (2\hat{i} - \hat{j} + 3\hat{k}) \\ &= 5\hat{i} - 2\hat{j} + 7\hat{k} \end{aligned}$$

$$\begin{aligned} \underline{BA} \cdot \underline{BC} &= (3\hat{i} + 4\hat{j} - \hat{k}) \cdot (5\hat{i} - 2\hat{j} + 7\hat{k}) \\ &= 15 - 8 - 7 = 0 \end{aligned}$$

Hence $\underline{BA}$ is perpendicular to $\underline{BC}$

$\therefore \angle ABC$ is a right angle.